The O365 Intrusion

1) What is the first timestamp at which a non-macOS device signed into Marcus Webb's account?
2) Just after logging in, the attacker accessed a large number of emails within the first few minutes of the session. How many emails were accessed during this window?
3) What subject line does the attacker's inbox rule match on?
4) How much time elapsed between the malicious inbox rule being created and a matching reply email arriving in the mailbox?
5) Marcus Webb's account logged a Teams message that had a link attached to it. What file did this link reference?
6) What operating system was used in the suspicious sign-in to David Chen's account?
7) What IP address was used in this same sign-in?
8) How much time passed between this message being created and the suspicious sign-in to David Chen's account?
9) The file the attacker downloaded from David Chen's account was legitimately modified earlier that day. At what time?
10) What file stored on the Leadership site had a sensitivity label applied to it by the organization's data governance, prior to the incident?
11) What file did the attacker create an external sharing link for?
12) What external email address was granted access to this file?
13) According to the emails, what is the subject line of the notification sent to this external address by the sharing application?

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1) What is the first timestamp at which a non-macOS device signed into Marcus Webb's account?

I'm building a search query using keywords, namely marcus and the login operation, to find operating systems other than macOS

1st Query: index=vantage marcus Operation=UserLoggedIn

Select Fields

Select All Within Filter | Deselect All | Coverage: 1% or more | Device | + Extract New Fields

i	Field	# of Values	Event Coverage	Type
>	DeviceProperties{}.Name	5	100%	String
v	DeviceProperties{}.Value	10	100%	String

Reports

Top values | Top values by time | Rare values | Events with this field

False	48	200%	
Chrome	24	100%	
MacOs	19	79.167%	
007e389a-c735-cd23-c1c1-929f755d54e0	8	33.333%	
007e389a-2615-eb4d-8934-82933ef67063	5	20.833%	
007e389a-ce37-f3d6-b0c5-59c28f4b5be7	5	20.833%	
Windows10	5	20.833%	
007e389a-cc3a-382d-0e86-3e50eb6460f5	3	12.5%	
007e389a-133e-e2f5-231c-2cc84ec0b2de	2	8.333%	
007e389a-b24c-98cd-5429-1a3576164e05	1	4.167%	

After finding a UserLoggedIn event with Windows OS, I use a subsequent query to identify the first login from that OS, while also excluding failed login attempts

2nd Query: index=vantage marcus Operation=UserLoggedIn "DeviceProperties{}.Value"=Windows10 ErrorNumber=0 | sort + _time


```
{ [-]
   Actor: [ [+]
   ]
   ActorContextId: 425c656e-fc78-4251-af5e-3db2e50cc196
   ActorIpAddress: 185.220.101.47
   ApplicationId: 9199bf20-a13f-4107-85dc-02114787ef48
   AzureActiveDirectoryEventType: 1
   ClientIP: 185.220.101.47
   CreationTime: 2026-08-27T12:28:31
   DeviceProperties: [ [+]
   ]
```


Answer Q1: 2026-08-27 12:28:31

2) **Just after logging in, the attacker accessed a large number of emails within the first few minutes of the session. How many emails were accessed during this window?**

To solve this question, I create a query covering the period after the first system login (identified in Q1), filter for email access, and sum OperationCount, as it represents the number of emails accessed

Query: index=vantage marcus.webb@vantagedynamics.onmicrosoft.com Operation=MailItemsAccessed | table _time, OperationCount | sort + _time

```
1 index=vantage marcus.webb@vantagedynamics.onmicrosoft.com Operation=MailItemsAccessed
2 | table _time, OperationCount
3 | sort + _time
```

Since date time

✓ **4 events** (8/27/26 12:28:31.000 PM to 10/12/26 1:29:45.000 AM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page Format Preview: On

_time	OperationCount
2026-08-27 12:28:40	6
2026-08-27 12:28:43	2
2026-08-27 12:28:53	8
2026-08-27 12:29:55	6

Answer Q2: 22

3) **What subject line does the attacker's inbox rule match on?**

To solve this, we need to examine the new inbox rules created by Marcus

Query: index=vantage marcus.webb@vantagedynamics.onmicrosoft.com Operation=New-InboxRule

```
AppId:
AppPoolName: Microsoft.Exchange.Clients.Owa2.Server.NetCore
ClientAppId:
ClientIP: 185.220.101.47:17747
CorrelationID:
CreationTime: 2026-08-27T12:30:10
ExternalAccess: false
Id: 4499a028-a1de-4c38-1b94-08df0436efde
ObjectId: nampr20a015.prod.outlook.com/Microsoft Exchange Hosted Organizations/vantagedynamics.onmicrosoft.com/b987ad11-7707-4dc5-ad19-dae6245e7a55\update
Operation: New-InboxRule
OrganizationId: 425c656e-fc78-4251-af5e-3db2e50cc196
OrganizationName: vantagedynamics.onmicrosoft.com
OriginatingServer: CY8PR20MB5378 (15.21.0360.006)
Parameters: [ [-]
  { [+]
  }
  { [+]
  }
  { [+]
  }
  { [-]
    Name: SubjectContainsWords
    Value: invoice
  }
  { [+]
  }
  { [+]
  }
]
RecordType: 1
RequestId: 2d54bf73-3a3e-97e1-05ca-cc726cac8716
```

Answer Q3: invoice

4) **How much time elapsed between the malicious inbox rule being created and a matching reply email arriving in the mailbox?**

Based on the question, we need to examine events in which Marcus receives emails within the time range after the inbox rule was created. (We could also add the keyword invoice to the query, since the inbox rule would not have worked without it. However, this is not necessary in this case)

Query: index=vantage RecipientAddress=marcus.webb@vantagedynamics.onmicrosoft.com | sort + _time



If the rule was created at 12:30:10 and the email appeared at 12:41:09, then 10 minutes and 59 seconds had elapsed

Answer Q4: 10 minutes 59 seconds

5) **Marcus Webb's account logged a Teams message that had a link attached to it. What file did this link reference?**

I decided to search using the keywords marcus.webb@vantagedynamics.onmicrosoft.com and Teams. This was sufficient to find the relevant event

Query: index=vantage marcus.webb@vantagedynamics.onmicrosoft.com Teams | table _time, ObjectId | sort + _time

2026-08-27 12:46:40	
2026-08-27 13:03:18	https://vantagedynamics.sharepoint.com/Shared Documents/Meridian_Invoice_Batch_Aug2026.pdf.pdf
2026-08-27 13:03:38	https://vantagedynamics.sharepoint.com/Shared Documents/Meridian_Invoice_Batch_Aug2026.pdf.pdf

Answer Q5: Meridian_Invoice_Batch_Aug2026.pdf

6) **What operating system was used in the suspicious sign-in to David Chen's account?**
7) **What IP address was used in this same sign-in?**

To answer these two questions, I decided to review David's Operation=UserLoggedIn events in chronological order to identify any deviations from his usual behavior

Query:index=vantage David Operation=UserLoggedIn ErrorNumber=0 | table _time, DeviceProperties{}.Value, ActorIpAddress | sort + _time

2026-08-27 09:24:57	MacOs Chrome False False 007e389a-9303-fdea-2ba3-4b2fbf5e0b85	41.128.55.10
2026-08-27 09:24:59	MacOs Chrome False False 007e389a-9303-fdea-2ba3-4b2fbf5e0b85	41.128.55.10
2026-08-27 13:27:04	Linux Chrome False False 007e389a-f9fa-c7c9-c3c5-1bfcd926a447	91.219.237.88
2026-08-27 13:27:04	Linux Chrome False 007e389a-f9fa-c7c9-c3c5-1bfcd926a447	91.219.237.88
2026-08-27 13:27:38	Linux Chrome False False 007e389a-f9fa-c7c9-c3c5-1bfcd926a447	91.219.237.88
2026-08-27 13:27:48	Linux Chrome False	91.219.237.88

We can see a login around the time of the compromise, as well as an unusual operating system and IP address

Answer Q6: Linux

Answer Q7: 91.219.237.88

8) **How much time passed between this message being created and the suspicious sign-in to David Chen's account?**

Here, we need to identify the exact time when the message containing Meridian_Invoice_Batch_Aug2026 was created

*Query: index=vantage Workload=MicrosoftTeams *Meridian_Invoice_Batch_Aug2026**

```
1  index=vantage Workload=MicrosoftTeams *Meridian_Invoice_Batch_Aug2026*
```

✓ **1 event** (before 10/12/26 2:08:13.000 AM) No Event Sampling

Events (1) Patterns Statistics Visualization

Timeline format — Zoom Out + Zoom to Selection × Deselect

> Show Fields Format Show: 20 Per Page View: List

i	Time	Event
>	8/27/26 1:03:38.000 PM	{ [-] AppAccessContext: { [+] } ChatThreadId: 19:b987ad11-7707-4dc5-ad19-dae6245e7a55_ce36 ClientIP: 185.220.101.47 CommunicationType: OneOnOne CreationTime: 2026-08-27T13:03:38 DisplayName: Marcus Webb ExtraProperties: [[+]

Meridian_Invoice_Batch_Aug2026 was created at 13:03:38, while David logged in at 13:27:04, as shown in the screenshots for Questions 6 and 7. Therefore, 23 minutes and 26 seconds elapsed between these two events.

Answer Q8: 23 minutes and 26 seconds

9) The file the attacker downloaded from David Chen's account was legitimately modified earlier that day. At what time?

Here, I decided to examine FileDownloaded events associated with the attacker's IP address and David's account

Query: index=vantage david Operation=FileDownloaded ClientIP=91.219.237.88 | table _time, ClientIP, SourceFileName

```
1  index=vantage david Operation=FileDownloaded ClientIP=91.219.237.88
2  | table _time, ClientIP, SourceFileName
```

✓ **1 event** (before 10/12/26 2:14:18.000 AM) No Event Sampling

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page Format Preview: On

_time	ClientIP	SourceFileName
2026-08-27 13:28:31	91.219.237.88	Q4_Board_Financial_Summary.xlsx

Answer Q9: Q4_Board_Financial_Summary.xlsx

10) What file did the attacker create an external sharing link for?

The question description did not specify which account initiated the sharing. I was also unsure of the exact Operation name, so I decided to create a query containing the attacker's IP address and an Operation field matching a keyword using wildcards

*Query: index=vantage 91.219.237.88 Operation=*Link* | table _time, Operation, ObjectId*

```
1  index=vantage 91.219.237.88 Operation=*Link*
2  | table _time, Operation, ObjectId
```

✓ **2 events** (before 10/12/26 2:17:23.000 AM) No Event Sampling Job

Events Patterns **Statistics (2)** Visualization

Show: 20 Per Page Format Preview: On

_time	Operation	ObjectId
2026-08-27 14:52:35	SharingLinkUpdated	https://vantagedynamics.sharepoint.com/Shared Documents/Vendor_Onboarding_Notes.docx.docx
2026-08-27 14:52:35	AnonymousLinkUpdated	https://vantagedynamics.sharepoint.com/Shared Documents/Vendor_Onboarding_Notes.docx.docx

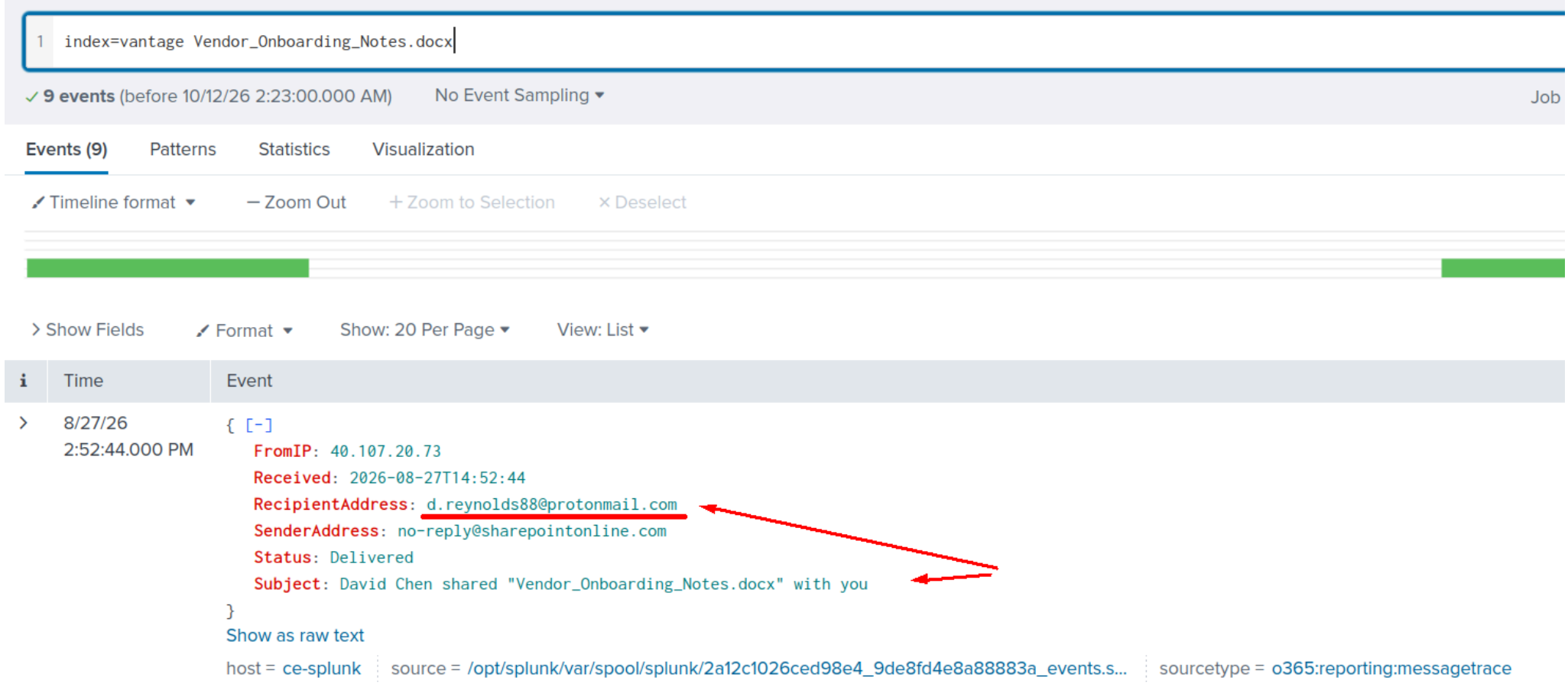
Answer Q10: Vendor_Onboarding_Notes.docx

11) What external email address was granted access to this file?

12) According to the emails, what is the subject line of the notification sent to this external address by the sharing application?

Simply added a keyword from the answer to Question 10 to the query and found the answers to Q11 and Q12

Query: index=vantage Vendor_Onboarding_Notes.docx



Honestly, I thought I was supposed to find this information in a different way, but this was the only log containing the keywords Vendor_Onboarding_Notes.docx and an email address, so this was probably the intended approach

Answer Q11: d.reynolds88@protonmail.com

Answer Q12: David Chen shared "Vendor_Onboarding_Notes.docx" with you